

MID-TERM EXAMINATION

GENERAL SCIENCE, MATHEMATICS & COMPUTING

Max Marks: 100

SECTION A: MATHEMATICS

1. A shopkeeper buys a bicycle for ₹2,400 and sells it at a profit of 15%. Find the selling price. [5 Marks]
8. Find the area of a right-angled triangle with base 12 cm and height 9 cm. [4 Marks]

SECTION B: BIOLOGY

2. Draw and label a diagram of a plant cell, showing the cell wall, nucleus, chloroplast, and vacuole. [8 Marks]
9. Draw a labeled diagram showing the process of photosynthesis, including the inputs and outputs. [8 Marks]

SECTION C: PHYSICS

3. (a) Derive the first equation of motion ($v = u + at$) using the definition of acceleration. [5 Marks]
- (b) A car accelerates from 5 m/s to 25 m/s in 4 seconds. Calculate its acceleration. [5 Marks]

SECTION D: GENERAL KNOWLEDGE

4. Which is the largest planet in our solar system? [2 Marks]
10. Who is known as the "Father of the Indian Constitution"? [2 Marks]

SECTION E: CHEMISTRY

5. (a) Draw the structure of methanal (HCHO) and name its functional group. [4 Marks]
- (b) Name the group and period of Sodium (Na) in the periodic table, and state one physical property. [4 Marks]

SECTION F: COMPUTER SCIENCE

6. Write a Python program to check whether a given number is prime. [6 Marks]

SECTION G: THEORY

7. (a) Explain the water cycle using a short story or real-life analogy. [5 Marks]
- (b) Explain Newton's Third Law using a short story or real-life analogy. [5 Marks]

[END OF QUESTION PAPER]